

MongoDB CRUD Assignment

INSERT (4 Questions)

1. Insert one employee document.

```
db.employees.insertOne({
  name: "John Smith",
  age: 30,
  department: "IT",
  salary: 40000
});
```

2. Insert multiple employee documents (at least 3).

```
db.employees.insertMany([
  { name: "Alice Johnson", age: 28, department: "HR", salary: 52000 },
  { name: "Bob Miller", age: 35, department: "IT", salary: 60000 },
  { name: "Carol White", age: 32, department: "Sales", salary: 45000 }
]);
```

3. Insert one product document.

```
db.products.insertOne({
  name: "Laptop",
  price: 850,
  category: "Electronics",
  stock: 25
});
```

4. Insert multiple order documents (at least 3).

```
db.orders.insertMany([
  { customer: "Raj", total: 1200, status: "Pending" },
  { customer: "Anita", total: 800, status: "Shipped" },
  { customer: "Karan", total: 650, status: "Delivered" }
]);
```

READ (6 Questions)

5. Retrieve all documents from employees collection.

```
db.employees.find();
```

6. Find one employee whose name is 'Alice Johnson'.

```
db.employees.findOne({ name: "Alice Johnson" });
```

7. Display only name and salary fields (exclude `_id`).

```
db.employees.find({}, { _id:0, name:1, salary:1 });
```

8. Find employees whose salary is greater than 50,000.

```
db.employees.find({ salary: { $gt: 50000 } });
```

9. Retrieve products whose price is between 500 and 1000.

```
db.products.find({ price: { $gte: 500, $lte: 1000 } });
```

10. Find employees in IT department and younger than 40.

```
db.employees.find({ department:"IT", age:{ $lt:40 } });
```

UPDATE (5 Questions)

11. Update salary of employee 'John Smith' to 45,000.

```
db.employees.updateOne(  
  { name: "John Smith" },  
  { $set: { salary: 45000 } }  
);
```

12. Increase salary by 5,000 for all employees in IT department.

```
db.employees.updateMany(  
  { department: "IT" },  
  { $inc: { salary: 5000 } }  
);
```

13. Update status of one order from Pending to Shipped.

```
db.orders.updateOne(  
  { status: "Pending" },  
  { $set: { status: "Shipped" } }  
);
```

14. Increase stock by 10 for all Electronics products.

```
db.products.updateMany(  
  { category: "Electronics" },  
  { $inc: { stock: 10 } }  
);
```

15. Change department from Sales to Marketing.

```
db.employees.updateMany(  
  { department: "Sales" },  
  { $set: { department: "Marketing" } }  
);
```

DELETE (5 Questions)

16. Delete one employee whose name is 'Carol White'.

```
db.employees.deleteOne({ name: "Carol White" });
```

17. Delete all employees whose salary is less than 40,000.

```
db.employees.deleteMany({ salary: { $lt: 40000 } });
```

18. Delete one order where status is Pending.

```
db.orders.deleteOne({ status: "Pending" });
```

19. Delete all orders where status is Delivered.

```
db.orders.deleteMany({ status: "Delivered" });
```

20. Delete all products where stock <= 0.

```
db.products.deleteMany({ stock: { $lte: 0 } });
```